

# One interface, four front ends

Each tinted box is one drive; the two left panels of a box are two different devices on that car. Badge “paper” marks the front end used in the paper’s experiments, “demo” the second device shown to make the interface point.

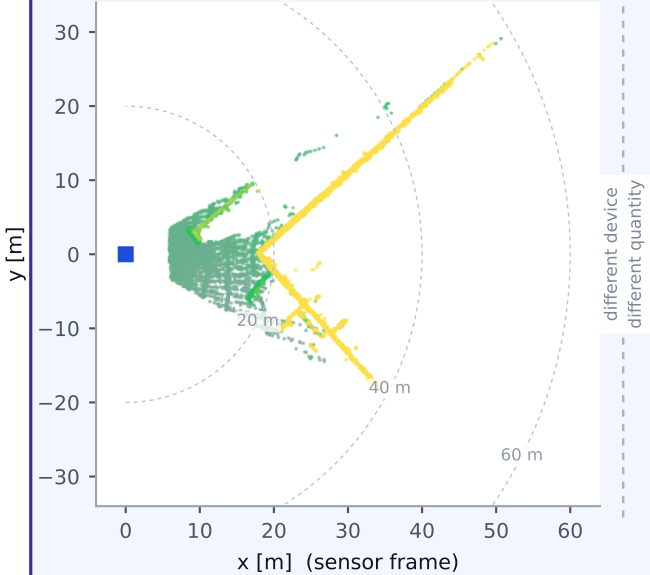
## A2D2 · drive 20180810\_150607

two independent devices · nothing shared: no field of view, no clock, no measured quantity

### SENSOR A · LiDAR

paper

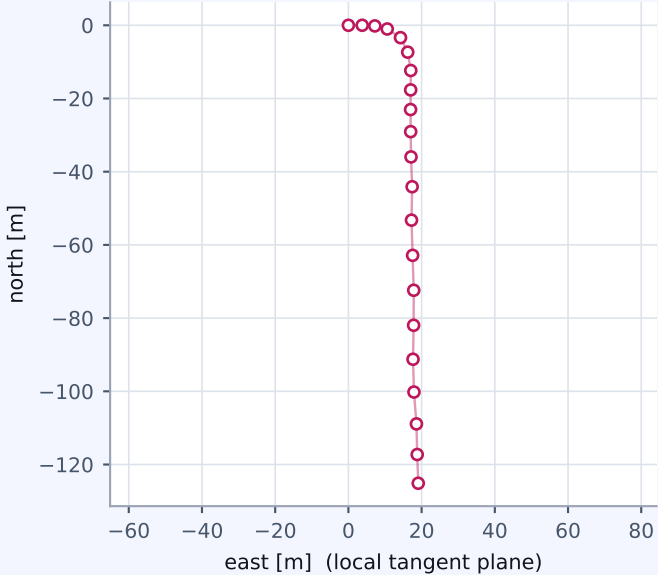
Velodyne VLP-16, FRONT\_CENTER view ·  $\approx 30$  Hz  
range + bearing to surfaces → scan matching  
shown: one scan, 14 000 points



### SENSOR B · GNSS

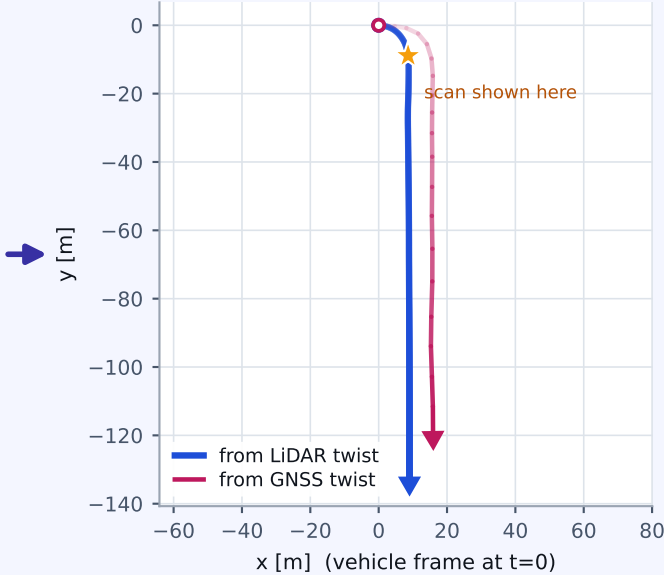
demo

vehicle-bus GNSS, position only · 1 Hz fix  
absolute position on Earth → differencing fixes  
shown: 21 fixes at 1.0 Hz, no attitude



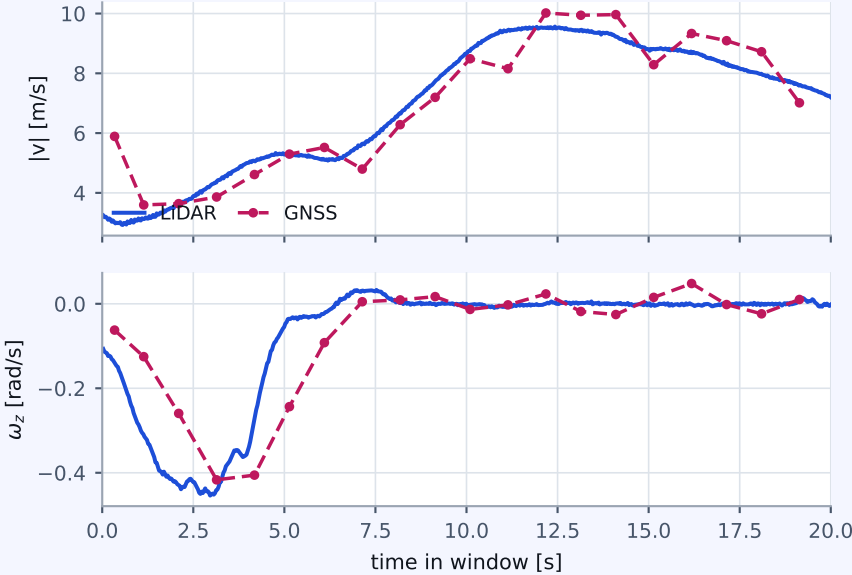
### Trajectory from ( $v$ , $\omega$ )

139 m driven,  $-88^\circ$  of heading



### The twist, as delivered

path speed 6.95 vs 6.98 m/s · median  $\omega$  gap 0.0242 rad/s



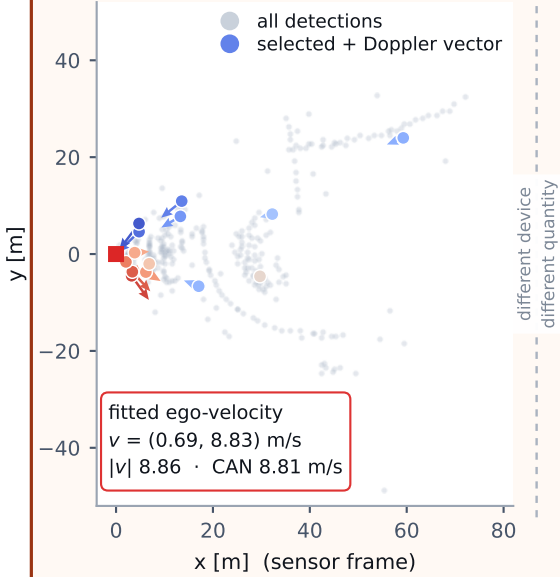
## RadarScenes · sequence 1

two independent devices · nothing shared: no field of view, no clock, no measured quantity

### SENSOR A · radar

paper

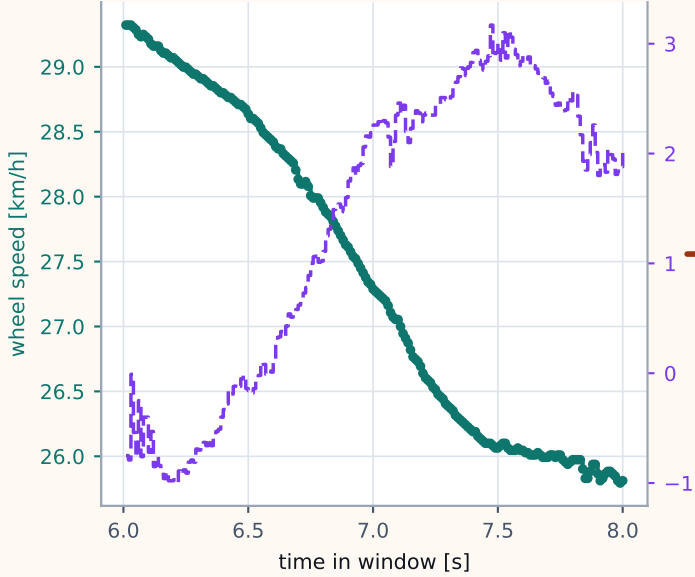
77 GHz series radar, sensor 1 · 13.5 scans/s  
range, azimuth, radial Doppler → one-scan least squares  
shown: one scan, 310 returns · 14 vectors



### SENSOR B · CAN bus

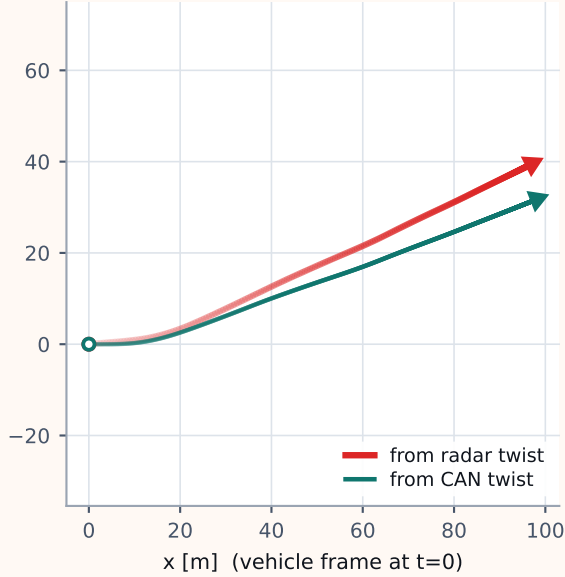
demo

wheel encoders + yaw-rate gyro · 100 Hz  
wheel rotation + yaw rate → vehicle kinematics  
shown: 2 s zoom of 2000 messages yaw rate [deg/s]



### Trajectory from ( $v$ , $\omega$ )

105 m driven,  $22^\circ$  of heading



### The twist, as delivered

median  $|v|$  gap 0.078 m/s · one 2-D radar gives no  $\omega$

